

E2 — Embedded Track, Week 1

Member: _____

Project setup (do first)

1. Fork the repo: <https://github.com/slvDev/esp32-ai>
2. Clone your fork. Add the original as upstream.
3. Read README.md, RESULTS.md, and research/tinystories/README.md.
4. Write 5 lines in the shared Drive doc: what you think the project does and where the hard part is.

Your task this week

Goal: week 5 measurements are defined before the board arrives.

Steps: 1. Build the bandwidth benchmark firmware on host. Compile only, no flash. See firmware/benchmarks/bandwidth. 2. Read the benchmark source. Write down what it measures and how. 3. Write the measurement protocol: tok/s, ms/token, memory free, flash row read, PSRAM and SRAM speeds. For each: the steps, the tool, the recording format, 3 repetitions.

Deliverable: MEASUREMENT_PROTOCOL.md in the club repo.

Verify: E1 reviews the protocol and finds no missing step.

Learning this week

Memory hierarchy. Explain why flash reads are slow and SRAM reads are fast. One sentence each.

Resources

- firmware/benchmarks/bandwidth: <https://github.com/slvDev/esp32-ai/tree/main/firmware/benchmarks/bandwidth>
- RESULTS.md (bandwidth table): <https://github.com/slvDev/esp32-ai/blob/main/RESULTS.md>
- ESP32-S3 docs: <https://docs.espressif.com/projects/esp-idf/en/latest/esp32s3/>

Done when

- [] MEASUREMENT_PROTOCOL.md exists in the club repo
- [] E1 finds no missing step in the protocol
- [] You can say what the project does in one sentence